

The Effects of Export Diversification and Concentration on Carbon Emissions: Asymmetric Evidence for Türkiye

Burcu Berke , Gülsüm Akarsu , Dilek Temiz 

Burcu Berke: Corresponding author, Professor, Nigde Ömer Halisdemir University, FEAS,
Department of Economics, Nigde/Türkiye, burcuberke@ohu.edu.tr

Gülsüm Akarsu: Associate Professor, Ondokuz Mayıs University, FEAS, Department of Political
Science and Public Administration, Samsun/Türkiye, gulsum.akarsu@omu.edu.tr

Dilek Temiz: Professor, Çankaya University, FEAS, Department of International Trade and Finance,
Ankara/Türkiye, dilektemiz@cankaya.edu.tr

Abstract

Numerous studies have explored the possible causes of global carbon emissions and climate change; however, the impact of export diversification and concentration on these emissions is a less examined topic in literature. This study investigates the effects of export diversification and concentration on carbon emissions in Türkiye from 1995 to 2018, utilizing the nonlinear ARDL method. The findings indicate that export diversification reduces carbon emissions after accounting for the effects of other control variables, such as renewable energy, levels of inequality, and government size. In this model, an increase in renewable energy, inequality, and government size leads to higher carbon emissions and worsens environmental quality, although the environmental Kuznets curve hypothesis is supported. Additionally, this study reveals that positive shocks in export concentration reduce carbon emissions, while negative shocks increase emissions. This hypothesis remains valid for the specified period in Türkiye, but the control variables display behavior similar to export diversification. The results suggest that mitigating climate change and improving environmental quality in Türkiye relies on policies that support export diversification and concentration.

Keywords: Export Diversification, Export Concentration, Carbon Emission, Türkiye, NARDL

Jel Codes: F140, Q56, C5

1. Introduction

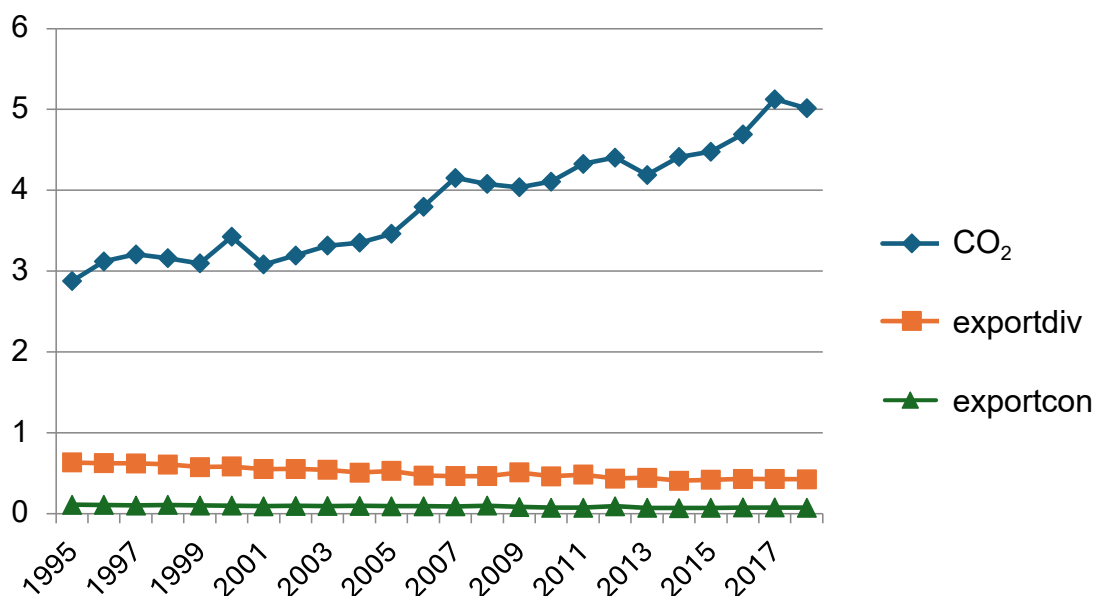
Beginning with Grossman and Krueger (1995), studies have explored the relationships between environmental degradation and various economic factors, such as trade, emphasizing economic growth and energy consumption (Wang et al., 2020). The connection between carbon emissions and economic growth was hypothesized as an inverted U-shaped relation known as the environmental Kuznets curve hypothesis, based on Kuznets (1955) (Cheng et al., 2019). This hypothesis indicates that carbon emissions increase as economies grow; however, emissions start to decline after reaching a certain level of economic growth as environmentally friendly technologies become more affordable (Rahman and Vu, 2020).

The export diversification strategy proposed by the IMF and World Bank aims to ensure the sustainable growth of developing countries, as the market share can increase through market diversification, product diversification, or both (Khan et al., 2021). However, while this strategy promotes increased production of export goods, it may also accelerate environmental degradation by leading to higher energy consumption. Export diversification can take two forms: market and product diversification. In the former, higher pollution levels can be expected due to the increased trading volume from more trading partners. In the latter, product diversification can be either vertical or horizontal (Iqbal et al., 2021). In vertical diversification, environmental damage can be mitigated through product improvements stemming from technological progress and increased value-added, which are related to learning by doing and technique effects. According to Murshed and Dao (2022), technological progress and R&D investments are crucial for improving export quality and reducing carbon emissions, as they enhance energy efficiency and promote the use of renewable energy. In the case of horizontal export product diversification, more significant environmental damage is likely due to the scale effect resulting from increased production of export goods.

According to the IPCC (2013), the increase in fossil fuel consumption since the 1970s has led to a gradual rise in carbon emissions and the effects of global warming, particularly in developing countries (Can et al. 2021). Extensive literature investigates the determinants of carbon emissions, and recent years have seen a growing discussion on the possible effects of foreign trade on these emissions. However, it is worth noting that the studies also examine export product diversification or concentration rather than solely focusing on foreign trade or openness. In other words, since introducing new products to the export product basket in developing countries can increase carbon emissions, the variety and concentration of export products can influence these emissions and the volume of foreign trade. Initially, the export product baskets of developing countries consist of traditional products, but their exports are

gradually diversified with the addition of more complex and information-intensive technological products. When this diversification surpasses a specific threshold value, the export concentration stage with limited pollution and manufactured products begins instead of pollution-intensive export diversification. Under these conditions, carbon emissions in developing countries are expected to rise as energy consumption is high during export product diversification, whereas carbon emissions are likely to decrease when manufactured products are limited in the export product concentration phase (Gözüör and Can 2016; Apergis et al., 2018; Zafar et al., 2021).

Figure 1: Trends in carbon emissions (CO₂), export diversification (exportdiv), and concentration (exportcon) in Türkiye



Source: UNCTAD and WDI

Figure 1 illustrates the trends in carbon emissions, export diversification, and concentration in Türkiye from 1995 to 2018. During this period, carbon emissions in the country rose gradually, while export diversification and concentration remained stable, with export diversification being slightly more significant than export concentration. In other words, the share of polluting or energy-intensive products within Türkiye's export basket was more substantial than that of information-intensive and sophisticated products.

In line with the Paris Agreement, Türkiye announced its 2053 net-zero emission goal in 2021. As a candidate country for the European Union and given that the EU is an important trade partner for Türkiye, the country published its Green Deal Action Plan in 2021 to contribute

to emission reductions and climate change mitigation. Transitioning to a low-carbon economy may provide numerous opportunities leading to higher economic growth, export diversification, and technological development (World Bank Group, 2022). Therefore, it is crucial for Türkiye to diversify its exports by producing sophisticated products to reduce carbon emissions and facilitate the transition to a low-carbon economy. Analyzing the case of Türkiye may yield insights into the effects of export diversification and concentration on carbon emissions, as Türkiye has adopted an export-led growth strategy since the 1980s. Thus, this study focuses on Türkiye.

The primary contribution of this study to the existing literature is to reveal the asymmetric effects of export diversification and concentration on carbon emissions in Türkiye during the period from 1995 to 2018. As is well known, there exist various quadratic or inverted U-shaped curves in the literature, with the most common being the examination of the Environmental Kuznets Curve (EKC) (Lind and Mehlum, 2010). The connections between carbon and energy are particularly prominent in existing literature (Cao et al., 2022). In this study, in addition to testing the validity of the EKC hypothesis, we will analyze the impact of renewable energy, income inequality, and government size on carbon emissions. In alignment with the Sustainable Development Goals, improving environmental quality or mitigating climate change rely on increasing the share of renewable energy sources along with implementing green finance and growth policies that promote such energy consumption. In a world where rising carbon emissions heighten concerns about climate change, it is crucial for investors to direct their funds toward renewable energy projects (Erdogan et al., 2024; Subramaniam and Loganathan, 2024).

Numerous studies in the literature examine the environmental impacts of various economic indicators. However, few inquire into the implications of export diversification and concentration on the environment or carbon emissions. This study aims to contribute to the literature by collectively analyzing the effects of export diversification and concentration on carbon emissions using the nonlinear ARDL method, while considering the influence of other control variables (growth, inequality, government size, and renewable energy) in Türkiye from 1995 to 2018.

2. Theoretical Ideas

This part of the study will present the theoretical expectations of the Environmental Kuznets Curve (EKC), export diversification and concentration, and other control variables used in the study, such as government size, inequality, and renewable energy concerning carbon emissions or environmental pollution.

- **Environmental Kuznets Curve:** This hypothesis suggests that pollution rises during the initial stages of development but declines once the per capita income surpasses a certain threshold. In these early stages, countries tend to prioritize job creation over environmental quality and focus heavily on industrialization. However, as income levels increase, so does environmental awareness, leading to a reduction in pollution as cleaner and greener technologies are adopted in the later stages of development. In essence, wealthier and more developed societies display institutional and non-institutional indicators that reflect a commitment to reducing environmental pollution once they cross a certain threshold. These societies increasingly appreciate the importance of a cleaner environment, making the less polluting service sector significant in their development (Grossman and Krueger, 1995; Clement and Meunie, 2010; Jayanthakumaran et al., 2012; Kaika and Zervas, 2013a, b.; Lau et al., 2014; Apergis, 2016; Gozgor and Can 2016, 2017; Walde-Rufael & Idawu, 2017; Apergis et al., 2018; Hailemariam et al., 2020; Lan et al., 2022).

From a sector-based perspective, the EKC hypothesis suggests that carbon emissions decrease once economic growth reaches a sufficient level or surpasses the threshold point for a country. In the early stages of economic growth, primary production is intense, natural resources are abundant, and waste production is high. Natural resources become depleted in the later stages of development and industrialization, while waste accumulates. There is a positive relationship between per capita income, or economic growth, and pollution during this period; however, the proliferation of services, technology, and information reduces waste in the economy as growth continues, leading to decreased carbon emissions (Kaika and Zervas, 2013a, p. 1393).

Although the EKC hypothesis predicts an inverted U-shaped relationship between carbon emissions and per capita income beyond a certain income level, this connection can shift into a positive N-shaped form, incorporating the effects of scale, composition, and technique. The scale effect contributes to environmental pollution, but the factors leading to pollution diminish in the composition and technical effects as cleaner technologies become increasingly preferred for production. In other words, the “scale” effect occurs when carbon emissions rise due to more inputs being used with heightened economic activity. In contrast, the “technical” effect refers to companies responding by gradually adopting cleaner production processes. The “composition” effect involves reducing the share of pollution-intensive goods in production. While the scale effect typically increases pollution, the composition effect arises when factor intensities significantly impact trade patterns. The two primary forces behind the EKC are structural changes and technological progress. Structural or compositional changes may include a shift in the production process from pollution-intensive industries to less polluting, knowledge-intensive service sectors.

Technological advancement improves production techniques, such as adopting greener technologies and/or minimizing inputs in the production process.

Increased production and depletion of natural resources lead to carbon emissions in the early stages of development. This phenomenon is known as the “scale” effect, which creates an upward trend along the Environmental Kuznets Curve (EKC) as production transitions from agriculture to industry. During this phase, economic development fosters investment in knowledge-based industries and the service sector, facilitating a shift toward cleaner technologies and improvements in production methods. These transitions are categorized as compositional and technical effects, respectively. A downward trend along the EKC curve is observed when both effects outweigh the scale effect. The compositional effect pertains to a change in production patterns from the more carbon-intensive manufacturing sector to a more sustainable service sector. In other words, environmental pollution initially arises from a shift in production composition, moving from light industry (agriculture or textiles) to heavy industry (minerals, chemicals, machinery, etc.). However, this is followed by a transition towards knowledge-intensive industries and services, which produce less pollution. The technical impact reflects a transition to cleaner technologies over older, polluting production methods and technological advancements, resulting in reduced input per capita of output. The development of cleaner techniques is promoted by investments in environmental research and development or by those that require a sufficient level of economic growth. Structural changes (compositional effect) and technological advancement (technical effect) emphasize the economic impact of production rather than consumption. Moreover, while the scale effect may be influenced by cyclical fluctuations, the compositional and technical effects require more time to manifest (Grossman and Krueger, 1995; Clement and Meunie, 2010; Jayanthakumaran et al., 2012; Kaika and Zervas 2013 a, b; Lau et al., 2014; Apergis, 2016; Gozgor and Can 2016, 2017; Hübler, 2017; Zhan et al., 2017; Apergis et al., 2018; Hailemariam et al., 2020).

- **Export diversification and concentration:** International trade also influences energy consumption and carbon emissions through three channels. Firstly, foreign trade and manufacturing increase energy consumption and carbon emissions due to the scale effect (where a rise in production boosts exports and imports). Secondly, developing countries can acquire high-impact manufacturing technology from developed countries (the “technical effect”) through foreign trade, leading to “less energy and more production.” Finally, as the economies of developing countries are agriculture-based, a structural transformation from agriculture to industry occurs. This diversification of products changes energy needs, resulting in a “composition effect.” In summary, foreign trade determines the increase or decrease in carbon emissions through these three effects.

Under these conditions, there is an inverted U-shaped relationship between a country's level of development and export diversification. Initially, developing countries expand their export product baskets by introducing new and diverse energy-intensive products during development. However, these countries reach a point where export product diversification diminishes as income levels rise. Beyond a certain threshold, developed countries shift towards product concentration in exports, focusing on complex products that require less energy. Energy consumption and carbon emissions are expected to be high during the export product diversification stage. In contrast, carbon emissions and pollution are anticipated to be low in the export product concentration stage. Regarding foreign trade theory, the Heckscher-Ohlin model suggests that a country with relatively low prices has abundant environmental resources. With greater openness, there is an increase in carbon emissions and pollution as specialization in energy-intensive products grows. Nonetheless, as the Stolper-Samuelson model takes effect, the costs that countries incur for the use of environmental resources rise, prompting companies to adopt fewer polluting methods (Ling et al., 2015, p. 20131; Can et al., 2021, pp. 33600–601).

- **Control variables:** Three arguments exist about how *income inequality* affects carbon emissions: political economy, marginal propensity, and the Veblen effect. The political economic argument posits that the wealthy derive disproportionate economic gains from polluting activities since they own polluting companies and depend on carbon-intensive industrial goods and services. Furthermore, the affluent can enjoy enhanced protection from the adverse effects of pollution due to their leverage in influencing policymakers, enabling them to evade substantial environmental costs. According to this argument, income inequality increases energy consumption, resulting in higher carbon emissions and shifting pollution costs onto low-income individuals. The “marginal propensity” argument builds on the Keynesian concept of consumption propensity. Income inequality fosters a greater marginal propensity to consume energy, as this margin is higher for low-income households than high-income households, leading to elevated carbon emissions. This indicates a negative relationship between inequality and carbon emissions. The final argument is the “Veblen or luxury consumption effect,” where competition for consumption occurs at the top of the income distribution, as individuals purchase goods and services to enhance their social status, thereby increasing carbon emissions and energy consumption. These arguments suggest that the effect of income inequality on carbon emissions can be either positive or negative. In summary, severe income inequality is likely to result in higher carbon emissions, while a certain level of income equality tends to reduce emissions (Ravallion et al. 2000, pp. 654, 655; Clement and Meunie, 2010, p. 419; Jorgenson et al. 2017, p. 41; Walde-Rufael and Idawu, 2017, p. 1339; Hailermariam et al. 2020, pp. 1142–43).

The impact of *government size* on carbon emissions is crucial in determining whether good governance enhances environmental quality. There are several channels through which government size can influence carbon emissions or environmental quality. First, a larger government size, particularly in developing countries, can positively affect environmental quality through economic growth. Conversely, government size can also detrimentally impact environmental quality, when the public sector utilizes resources inefficiently or when public investments and subsidies in agriculture, industry, and energy increase carbon emissions. Moreover, the government plays a proactive role in environmental protection. In these contexts, governments can steer societies toward energy-efficient products and green technologies, suggesting that increasing government size can further enhance environmental quality. Besides these channels, the effects of government size on the environment may be direct or indirect. In the direct effect, income equality improves, and environmental quality rises as public expenditures incorporate more redistributive transfers. In the indirect effect, the overall impact of government size on the environment remains uncertain, as its effect on environmental pollution can be either low or high, depending on the nature of the Environmental Kuznets Curve (Halkos and Paizanos, 2013, p. 48; Lan et al. 2022, p. 47226). Considering the impact of *renewable energy* on carbon emissions, increasing the use of renewables can reduce carbon emissions. In other words, renewable energy decreases both carbon emissions and pollution, making it preferable to non-renewables (natural gas, oil, and coal) (Bilgili et al., 2016, p. 841; Akadiri and Adebayo, 2022, p. 16314).

3. Selected Empirical Studies

Early studies on this subject focus on the EKC hypothesis. This indicates that carbon emissions increase up to a certain threshold as per capita income rises in an economy. However, emissions gradually decline, and environmental quality improves as economic growth continues (Grossman & Krueger, 1995). Many empirical studies support the EKC hypothesis, including those by Ang (2007), Tamazian et al. (2009), Clement and Meunie (2010), Jalil and Feridun (2011), Iwata et al. (2012), Jayanthakumaran et al. (2012), Öztürk and Acaravcı (2013), Shahbaz et al. (2013b), Lau et al. (2014), Ling et al. (2015), Apergis (2016), Bilgili et al. (2016), Doğan and Şeker (2016), Zhan et al. (2017), Haug and Ucal (2019), and Nkwatoh (2022).

The following part of the study features a review of selected literature. First, empirical studies that find export diversification and concentration increase carbon emissions will be examined. This will be followed by empirical studies suggesting that the relevant variables reduce these emissions. Finally, studies will test the effects of other control variables (income inequality, government size, and renewable energy) on carbon emissions.

(i) Studies examining the effects of export diversification on environmental pollution by Adewuyi and Awodumi (2016) for Nigeria from 1981 to 2014; Gözgör and Can (2016) for Türkiye from 1971 to 2010; Adamu et al. (2019) for India from 1983 to 2014; Fang et al. (2019) for 82 developing countries from 1970 to 2017; Can et al. (2020) for 84 developing countries from 1971 to 2014; Mania (2020) for 98 countries from 1995 to 2013; Wang et al. (2020) for G7 countries from 1990 to 2017; Can et al. (2021) for ten recently industrialized countries from 1970 to 2014; Iqbal et al. (2021) for 37 OECD countries from 1970 to 2019; and Khan et al. (2021) for 12 countries from 1987 to 2017, suggest that product diversification in the long term increases carbon emissions.

(ii) Apergis et al. (2018) found that carbon emissions decreased for 19 developed countries during the 1962–2010 period, alongside an increase in export product concentration (or the diversification of the product basket), as these countries produced those products. Studies by Liu et al. (2019) covering 125 countries in the 2000–2015 period and Shahzad et al. (2020) analyzing 63 countries from 1971 to 2014 indicate that export diversification reduced carbon emissions or negatively impacted environmental pollution. Zafar et al. (2021) observed that export diversity decreased carbon emissions in 22 countries with the highest immigration levels from 1986 to 2017. Conversely, Sharma et al. (2021) discovered that carbon emissions rose as export diversification declined or export product concentration increased in BRICS countries during the 1990–2018 period.

Several studies also investigate the effects of export product quality on carbon emissions. Gözgör and Can (2017) found that carbon emissions decreased as export product quality increased in China during the 1971–2010 period. In contrast, studies by Wang et al. (2021) for ten countries in the 1980–2014 period and Murshed and Dao (2022) for five South Asian countries during the 1972–2014 period found that export product quality was linked to increased carbon emissions.

(iii) Selected studies analyzing the effects of income inequality on carbon emissions by Baek and Gweisah (2013) for the USA during the 1967–2008 period; Zhang and Zhao (2014) for 28 Chinese provinces during the 1995–2010 period; Jorgenson (2015) for selected OECD member and non- member countries during the 1990–2008 period; Haoa et al. (2016) for 23 Chinese provinces during the 1995–2012 period; Jorgenson et al. (2017) for the USA during the 1997–2012 period; Walde- Rufael and Idawu (2017) for China during the 1974–2010 period and for India during the 1971–2010 period; Demir et al. (2019) for Türkiye during the 1963–2011 period; Yameogo and Dauda (2020) for Burkina Faso and Nigeria during the 1980–2016 period; Ali (2021) for Egypt during the 1975–2017 period; Sharma et al. (2021) for BRICS

countries during the 1990–2018 period; Baloch and Danish (2022) for BRICS countries during the 1994–2018 period; Yang et al. (2022) for 42 countries during the 1984–2016 period; Wang et al. (2023a) for 139 countries during the 1998–2018 period; and Wang et al. (2023b) for 56 countries during the 2003–2018 period, all of which found that income inequality increased environmental pollution. In contrast to these studies, Ravallion et al. (2000), Hübler (2017), and Andersson (2023) found that income inequality reduces carbon emissions. Finally, Clement and Meunie (2010) discovered that income inequality reduced carbon emissions in transition countries but increased emissions in developing countries across 83 countries during the 1988–2003 period. In their study on 17 OECD countries during the 1945–2010 period, Hailemariam et al. (2020) stated that carbon emissions increased as the share of income received by the wealthiest 10% of the population rose, while these emissions decreased and environmental quality improved as income inequality, measured by the Gini coefficient, increased. In their study on G7 countries during the 1870–2014 period, Uddin et al. (2020) found that carbon emissions increased in parallel with income inequality between 1870 and 1880. However, emissions decreased as this inequality grew during the 1950–2000 period.

Ajide and İbrahim's (2022) study, which is based on four different income inequality indicators (Gini coefficient, Palma ratio, Theil index, and Atkinson index) for G7 countries during the 1971–2015 period, along with Safar's (2022) study using two income inequality indicators (Gini coefficient and Atkinson index) for France during the 1980–2018 period, reported that the impact of inequality on carbon emissions varies by country. However, Safar (2022) found that net inequality indicators reduce carbon emissions, consistent with the marginal emissions propensity theory.

(iv) Among the selected studies analyzing the effects of fiscal policy on carbon emissions, Halkos and Paizanos (2013) indicated that government spending reduced environmental pollution across 77 countries during the 1980–2000 period. In another study, Halkos and Paizanos (2016) discovered that a phase of expansionist fiscal policy lowered carbon emissions from production and consumption in the USA throughout the 1973–2013 period. Research conducted on 106 Chinese cities from 2002 to 2014 by Zhan et al. (2017) and on France during the 1990–2018 period by Azam et al. (2023) showed that carbon emissions decreased as government size increased. Yameogo and Dauda (2020) found that government spending in Burkina Faso and Nigeria from 1980 to 2016 led to a long-term reduction in carbon emissions. Conversely, unlike the aforementioned studies, Katircioğlu and Katircioğlu (2018) asserted that expansionary fiscal policy in Türkiye during the 1960–2013 period resulted in increased carbon emissions. Chen's (2022) research on BRICS countries from 1990 to 2018 and Lan et al.'s (2022) study on China during the 1971–2019 period indicated that carbon emissions rose alongside increased government size.

(v) Selected studies analyzing the effects of renewable energy on carbon emissions by Baek (2016) for the USA during the 1960–2010 period; Bilgili et al. (2016) for 17 OECD countries during the 1977–2010 period; Doğan and Şeker (2016) for EU countries during the 1980–2012 period; Cheng et al. (2019) for BRICS countries during the 2000–2013 period; Mahmood et al. (2019) for Pakistan during the 1980–2014 period; Rahman and Vu (2020) for Australia and Canada during the 1960–2015 period; Namahoro et al. (2021) for 50 African countries during the 1980–2018 period; Zafar et al. (2021) for 22 countries receiving the most immigration during the 1986–2017 period; Wang et al. (2021) for 10 countries during the 1980–2014 period; Adebayo et al. (2022a) for MINT countries during the 1990–2018 period; Adebayo et al. (2022b) for Sweden during the 1965–2019 period; Akadiri and Adebayo (2022) for India during the 1970–2018 period; Baloch and Danish (2022) for BRICS countries during the 1994–2018 period; Sun et al. (2022) for Middle Eastern and North African countries during the 1991–2019 period; Wang (2022) for China during the 2007–2019 period; Adebayo et al. (2023) for Türkiye during the 1965–2019 period; Khan et al. (2023) for Belt and Road countries during the 2002–2019 period; and Raihan et al. (2022) for Malaysia during the 1990–2019 period, all found that renewable energy reduced carbon emissions.

4. Empirical Section: Data, Methods, Models, and Empirical Results

The effects of export diversification and concentration on carbon emissions in Türkiye from 1995 to 2018 were examined using the NARDL method after assessing the influence of other control variables outlined in this section of the study.

4.1 Data

The purpose of this study is to investigate the effects of export product diversification and concentration on carbon emissions in Türkiye during the 1995–2018 period, after accounting for the impact of other control variables such as income inequality, government size, growth, and renewable energy, using the nonlinear ARDL method. “A negative shock (decrease) or positive shock (increase) in one variable may not indicate a corresponding change in other economic variables. Therefore, various factors can create asymmetric relationships among macroeconomic variables. An asymmetric relationship between the relevant variables indicates that negative or positive shocks in the independent variables produce unrelated effects on the dependent variable” (Akadiri and Adebayo, 2022, p. 16312). After evaluating other specified control variables, this study will reveal the asymmetric effects of export diversification

and concentration on carbon emissions. The series for export product diversification and concentration were obtained from *UNCTAD*, the Gini coefficient, used as the primary indicator of income inequality, was sourced from *SWIID*, and the remaining variables were taken from *WDI*. The study illustrates the asymmetric effects (positive and negative shocks) of export diversification and concentration on carbon emissions based on two models.

4.2 Method

This study uses the nonlinear ARDL estimation method to demonstrate the asymmetric effects of export diversification and export concentration ($\ln export$) along with other control variables (economic growth ($\ln gdp$), squared economic growth ($\ln gdp^2$), renewable energy ($\ln renewable$), income inequality ($\ln gini$), and government size ($\ln govcons$)) on carbon emissions ($\ln CO_2$). Shin et al. (2014) developed this method to allow for short- and long-run asymmetries. A nonlinear structure is achieved by integrating the positive and negative partial sum decompositions of the independent variables in both the short and long run. This nonlinear model can be estimated using OLS. Cointegration tests can be conducted based on the bounds testing approach, allowing mixed integration orders. Impulse-response functions can be derived separately for positive and negative shocks.

$$\ln CO_2 = f(\ln export, \ln gdp, \ln gdp^2, \ln gini, \ln govcons, \ln renewable) \quad (1)$$

For the model presented in equation (1), the asymmetric long-run relationship can be illustrated as shown in equation (2).

$$\begin{aligned} \ln CO_{2t} &= x_t^+ \beta^+ + x_t^- \beta^- + \varepsilon_t \\ \Delta x_t &= \epsilon_t \end{aligned} \quad (2)$$

Where x is a 6×1 vector of explanatory variables, including export diversification (export concentration), renewable energy consumption, income inequality, government size, GDP per capita, and squared GDP per capita. Equation (3) shows the decomposition of x_t , and in the equation, x_t^+ and x_t^- are partial sum processes of positive and negative changes in x_t .

$$x_t = x_t^+ + x_t^- + x_0 \quad (3)$$

where, $x_t^+ = \sum_{k=1}^t \Delta x_k^+ = \sum_{k=1}^t \max(0, \Delta x_k)$ and $x_t^- = \sum_{k=1}^t \Delta x_k^- = \sum_{k=1}^t \min(0, \Delta x_k)$.

The nonlinear ARDL (p, q) model can be illustrated in equation (4).

$$\ln CO_{2t} = \sum_{i=1}^p \alpha_i \ln CO_{2t-i} + \sum_{i=0}^q (x_{t-i}^+ \beta_i^+ + x_{t-i}^- \beta_i^-) + u_t \quad (4)$$

Here, $u_t \sim iid(0, \sigma_u^2)$, α_i , β_i^+ , and β_i^- represent error term, autoregressive, and asymmetric distributed lag parameters, respectively. Equation (4) can be written in the error correction form as in equation (5);

$$\begin{aligned} \Delta \ln CO_{2t} = & \rho \ln CO_{2t-1} + x_{t-1}^+ \beta^+ + x_{t-1}^- \beta^- + \sum_{i=1}^{p-1} \delta_i \Delta \ln CO_{2t-i} + \\ & + \sum_{i=0}^{q-1} (\Delta x_{t-i}^+ g_i^+ + \Delta x_{t-i}^- g_i^-) + u_t \end{aligned} \quad (5)$$

In equation (5), $\rho = \sum_{i=1}^p \alpha_i - 1$; $\delta_i = -\sum_{j=i+1}^p \alpha_j$, $i = 1, \dots, p-1$; $\beta^+ = \sum_{i=0}^q \beta_i^+$; $\beta^- = -\sum_{i=0}^q \beta_i^-$; $g_0^+ = \beta_0^+$; $g_i^+ = -\sum_{j=i+1}^q \beta_j^+$, $i = 1, \dots, q-1$; $g_0^- = \beta_0^-$; $g_i^- = -\sum_{j=i+1}^q \beta_j^-$, $i = 1, \dots, q-1$.

The nonlinear error correction term is $ecm_t = \ln CO_{2t} - x_t^+ \theta^+ - x_t^- \theta^-$ where, $\theta^+ = -\beta^+/\rho$ and $\theta^- = -\beta^-/\rho$ are asymmetric long-run parameters.

The OLS estimation method is employed to estimate the nonlinear ARDL model. Symmetricity in the long run and short run coefficients can be tested using the Wald statistic with a χ^2 asymptotic distribution. By following the bounds testing approach of Pesaran, Shin, and Smith (2001) to test the existence of long-run relationships, t_{BDM} and F_{PSS} tests were proposed by Shin et al. (2014) to test $H_0 : \rho = 0$ against $H_1 : \rho < 0$ and $H_0 : \rho = \beta^+ = \beta^- = 0$ jointly, respectively.

Cumulative dynamic multiplier effects of x_t^+ and x_t^- on $\ln CO_{2t}$ can be obtained by the following expressions in equations (6) and (7), respectively.

$$m_h^+ = \sum_{i=0}^h \frac{\partial \ln CO_{2t+i}}{\partial x_t^+} \quad (6)$$

$$m_h^- = \sum_{i=0}^h \frac{\partial \ln CO_{2t+i}}{\partial x_t^-} \quad (7)$$

where, $h = 0, 1, 2, \dots$ and as $h \rightarrow \infty$, $m_h^+ \rightarrow \theta^+$ and $m_h^- \rightarrow \theta^-$. The lag lengths were selected based on diagnostic tests and information criteria.

4.3 Model and Empirical Results

All variables in the study are modeled in natural logarithmic form, and equation (8) below is estimated using the nonlinear ARDL method:

$$\ln CO_t^2 = \alpha_0 + \beta_1 \ln export_t + \beta_2 \ln gdppc_t + \beta_3 \ln gdppc_t^2 + \beta_4 \ln gini_t + \beta_5 \ln govcons_t + \beta_6 \ln renewable_t + \varepsilon_t \quad (8)$$

In which CO^2 represents carbon emissions, *export* shows export diversification or concentration, *gdppc* indicates GDP per capita, $gdppc_t^2$ shows the square of GDP per capita, *gini* represents the Gini coefficient used to indicate income inequality, *govcons* indicates the ratio of government spending to GDP, *renewable* represents renewable energy consumption, *t* shows the time dimension, and ε indicates the error term. Consistent with the theoretical literature, the β_1 coefficient is expected to be positive for export product diversification but negative for export product concentration. Consistent with the EKC hypothesis, β_2 and β_3 are expected to be positive and negative, respectively. In other words, according to this hypothesis, carbon emissions initially increase parallel to a rise in per capita income. Still, emissions decrease after a particular threshold value as the income level improves (Grossman and Krueger, 1995). On the other hand, the β_4 , β_5 and β_6 coefficients could be either positive or negative based on theoretical expectations.

To apply the NARDL method in the study, the dependent variable must be stationary at the first difference [I (1)], and it should be confirmed that no variable in the model is stationary at the second difference [I (2)]. Therefore, the Phillips-Perron (PP, 1988) unit root test was conducted for the relevant variables, as shown in Table 1, and the KPSS (1992) unit root test was performed for the level and squared GDP per capita variables that exhibited exploding roots. The results indicated that all variables in the model were stationary at their first differences. In other words, based on the PP (1988) unit root test findings, the condition of being stationary at the first difference for logarithmic carbon emissions, the dependent variable for the NARDL estimation, is satisfied. It has also been verified that none of the variables in the model are stationary at the second difference, considering the possibility that the F statistics are insignificant. The NARDL estimation commenced after confirming that the degree of integration of the variables in the model is maximum first-order integrated [I (1)].

Table 1: Results from PP (1988) and KPSS (1992)

variable	statistics	variable	statistics
$\ln CO_2$	-0.433761	$\Delta \ln CO_2$	-6.313875***
$\ln export$	-1.211332	$\Delta \ln export$	-8.555638***
$\ln gdppc$	0.691951	$\Delta \ln gdppc$	0.091713***
$\ln gdppc^2$	0.691196	$\Delta \ln gdppc^2$	0.102326***
$\ln gini$	-2.116655	$\Delta \ln gini$	-3.785269***
$\ln govcons$	-2.242736	$\Delta \ln govcons$	-8.612344***
$\ln renewable$	-1.517237	$\Delta \ln renewable$	-8.209289***
$\ln expcons$	-1.526317	$\Delta \ln expcons$	-12.18853***

Notes: "****" shows the value of 1% probability, and "Δ" shows the first difference. BDS tests were performed for both models and results show evidence of nonlinearity. Moreover, results of Oster (2019) delta test show absence of any omitted variable bias using psacalc2 Stata command. Results are available on request.

Source: Based on the authors' own calculations.

In the study, Equation (8) can be expressed in ARDL form as follows:

$$\begin{aligned}
 \Delta \ln CO_{2t} = & \alpha_0 + \sum_{i=1}^n \mu_1 \Delta \ln CO_{2t-i} + \sum_{i=0}^n \mu_2 \Delta \ln export_{t-i} + \sum_{i=0}^n \mu_3 \Delta \ln gdppc_{t-i} \\
 & + \sum_{i=0}^n \mu_4 \Delta \ln gdppc^2_{t-i} + \sum_{i=0}^n \mu_5 \Delta \ln gini_{t-i} + \sum_{i=0}^n \mu_6 \Delta \ln govcons_{t-i} \\
 & + \sum_{i=0}^n \mu_7 \Delta \ln renewable_{t-i} + \gamma_0 \ln CO_{2t-1} + \gamma_1 \ln export_{t-1} + \gamma_2 \ln gdppc_{t-1} \\
 & + \gamma_3 \ln gdppc^2_{t-1} + \gamma_4 \ln gini_{t-1} + \gamma_5 \ln govcons_{t-1} + \gamma_6 \ln renewable_{t-1} \\
 & + \alpha ECM_{t-1} + \omega_t
 \end{aligned} \tag{9}$$

In equation (9), n is the optimum delay length, Δ is the difference operator, γ_0 to γ_6 is the long-term relationship between the variables, and μ_1 to μ_7 are the total signs corresponding to the short-term dynamics of the variables. On the other hand, α_0 is the constant term, ω_t is the error term, and α is the error correction mechanism coefficient.

The model specification is chosen as an unconstrained constant with no trend, and Equation (9) is estimated using the Akaike information criterion. Equation (9) establishes the long-term relationship to test Pesaran et al. (2001)'s bound procedures. The null hypothesis of the bound procedure is $H_0: \gamma_0 - \gamma_1 - \gamma_2 - \gamma_3 - \gamma_4 = \gamma_5 = \gamma_6 = 0$, which means the absence of a long-run relationship or cointegration. The alternative hypothesis is $H_A: \gamma_0 \neq 0, \gamma_1 \neq 0, \gamma_2 \neq 0, \gamma_3 \neq 0, \gamma_4 \neq 0, \gamma_5 \neq 0, \gamma_6 \neq 0$ and indicates a long-term relationship or cointegration. The joint significance is tested with an F test in Equation (9). This F test has a non-standard distribution. Pesaran et al. (2001) calculated two asymptotic critical values, and the acceptance or rejection of the hypothesis is based on comparing the F statistic with these critical values. Narayan (2005) was employed to assess cointegration in the study due to the small sample size. When the calculated F statistic exceeds the upper limit of the critical value [I (1)], a long-term cointegration relationship between the variables, specifically between carbon emissions and macroeconomic factors, emerges.

Estimating a long-term relationship through the cointegration test in the linear or symmetric ARDL approach relies on the assumption of symmetry, where the explanatory variables impact the dependent variable linearly. However, since variations in a variable can occur in positive and negative directions, it is crucial to account for these changes in the independent variable. The asymmetric relationship between variables is explored further using the nonlinear or asymmetric ARDL method developed by Shin et al. (2014). This study constructs a nonlinear or asymmetric ARDL model based on the linear or symmetric ARDL equations (10). Logarithmic efficiency is divided into positive and negative changes among the independent variables, represented as $\ln export+$ and $\ln export-$.

$$\begin{aligned}
 POS(export)_t &= \sum_{L=1}^t \ln export_L^+ = \sum_{L=1}^T MAX(\Delta \ln export_L, 0) \\
 NEG(export)_t &= \sum_{L=1}^t \ln export_L^- = \sum_{L=1}^T MIN(\Delta \ln export_L^-, 0)
 \end{aligned} \tag{10}$$

Equation (9) can be expressed in asymmetric or nonlinear ARDL (NARDL) format as follows, incorporating negative and positive shocks in export diversification or concentration:

$$\begin{aligned}
 \Delta \ln CO_{2t} = & \alpha_0 + \sum_{i=1}^n \mu_1 \Delta \ln CO_{2t-i} + \sum_{i=0}^n \mu_2^+ \Delta \ln POS(export)_{t-i} + \\
 & + \sum_{i=0}^n \mu_2^- \Delta \ln NEG(export)_{t-i} + \sum_{i=0}^n \mu_3^- \Delta \ln gdppc_{t-i} + \sum_{i=0}^n \mu_4 \Delta \ln gdppc^2_{t-i} + \sum_{i=0}^n \mu_5 \Delta \ln gini_{t-i} + \\
 & + \sum_{i=0}^n \mu_6 \Delta \ln govcons_{t-i} + \sum_{i=0}^n \mu_7 \Delta \ln renewable_{t-i} + \gamma_0 \ln CO_{2t-1} + \gamma_1^+ \ln POS(export)_{t-1} + \\
 & + \gamma_1^- \ln NEG(export)_{t-1} + \gamma_2 \ln gdppc_{t-1} + \gamma_3 \ln gdppc^2_{t-1} + \gamma_4 \ln gini_{t-1} + \\
 & + \gamma_5 \ln govcons_{t-1} + \gamma_6 \ln renewable_{t-1} + \omega_t
 \end{aligned} \tag{11}$$

According to Shin et al. (2014), long-term cointegration is confirmed using the bound test approach, which involves comparing the F statistic to the critical value, as Pesaran et al. (2001) suggested. The null hypothesis affirms the existence of long-run cointegration between carbon emissions and explanatory variables. Additionally, the Wald test examines whether an asymmetric relationship exists between the dependent and explanatory variables. As shown in Table 2, the W_{LR} test displays the Wald test statistics for long-term symmetry, while the W_{SR} test presents the Wald statistic for short-term symmetry. Reviewing the results of Models 1 and 2, the null hypothesis of a symmetric relationship for the long term is rejected at a 1% significance level. Therefore, the relevant variables exhibit an asymmetric relationship in the long run. The null hypothesis regarding symmetric relationships is also rejected at the 1% significance level for the short term. These findings indicate an asymmetric relationship between carbon emissions and the independent variables (export diversification or concentration, GDP per capita level and square, inequality, government size, and renewable energy) in Türkiye during the 1995-2018 period.

Table 2: Results of the Cointegration Tests

Model	F Statistics	1% CV		5% CV	
		I(0)	I(1)	I(0)	I(1)
1	6.969755	4.77	6.67	3.35	4.77
2	27.07032	4.77	6.67	3.35	4.77

Source: Narayan (2005); CV indicates a critical value.

Examining the long-term estimation results for Model 1 in Table 3, which includes the effects of export diversification on carbon emissions, a positive shock in export diversification is associated with a decrease in carbon emissions in Türkiye during the relevant period, reducing emissions by 0.91%, while a negative shock similarly decreases emissions by 1.48%. Under these circumstances, both positive and negative shocks in export diversification over the long term in Türkiye from 1995 to 2018 have contributed to reduced carbon emissions and improved environmental quality. An analysis of the effects of other variables on carbon emissions indicates that while the logarithmic GDP per capita has a positive impact in the long run, the EKC or inverted U-shaped hypothesis holds true as its square shows a negative relationship. Conversely, 1% increases in income inequality, government size, and renewable energy are associated with increased carbon emissions by 2.15%, 0.49%, and 0.83%, respectively. Regarding the short-term nonlinear ARDL results for Model 1, it appears that, contrary to the long-term effects, one and two lag negative shocks in export diversification led to increases in carbon emissions by 2% and 1.24%, respectively. Furthermore, the EKC relationship is U-shaped since both the level and square of the logarithmic GDP per capita are negative and positive in the short term. Unlike the long term, increased income inequality in the short term reduces carbon emissions by 2.04%.

Table 3: Estimation Results

Model 1			Model 2	
Variables	Coefficient	T statistics	Coefficient	T statistics
$\ln\text{CO}_2$	-0.677352	-1.8072	-2.839335	-6.3722***
$\ln\text{export_neg}$	-1.482446	-3.4205**		
$\ln\text{export_pos}$	-0.919772	-3.0535**		
$\ln\text{expcons_neg}$			0.365055	2.7808*
$\ln\text{expcons_pos}$			-0.311719	-2.3984*
$\ln\text{gdppc}$	62.55556	5.1532***	21.03864	2.4642*
$\ln\text{gdppc}^2$	-3.470246	-5.0611***	-1.039652	-2.1703
$\ln\text{gini}$	2.154390	2.5492*	0.985504	2.0681
$\ln\text{govcons}$	0.496987	2.5234*	2.345560	6.8294**
$\ln\text{renewable}$	0.837345	5.5714*	0.265267	3.5222**
$\Delta\ln\text{gini}(-1)$	-2.045696	-3.5784**		
$\Delta\ln\text{export_neg}(-1)$	2.001099	4.1899**		
$\Delta\ln\text{export_neg}(-2)$	1.242082	4.1883**		
$\Delta\ln\text{expcons_neg}(-1)$			0.469060	4.7286**
$\Delta\ln\text{expcons_neg}(-2)$			0.341595	3.3613**
$\Delta\ln\text{gdppc}(-1)$	-99.28457	-6.1336***	-41.16186	-3.8725**
$\Delta\ln\text{gdppc}^2(-1)$	5.533060	6.1264***	2.232647	3.8222**
$\Delta\ln\text{CO}_2(-1)$	0.904539	2.7675*	0.579269	1.8313
$\Delta\ln\text{CO}_2(-2)$	-0.488780	-2.2121*	-0.273805	-1.4953
$\Delta\ln\text{govcons}(-1)$			-1.003937	-4.5117**
$\Delta\ln\text{renewable}(-1)$			-0.203687	-2.5775*
R^2		0.9784		0.9919
F		12.1347**		23.0547**
DW		2.72		3.04

Notes: "Δ" indicates the first difference of the variables. *, **, *** signs indicate 10%, 5% and 1% significance levels, respectively.

Source: Based on the authors' own calculations.

Examining the results of the diagnostic tests related to the nonlinear ARDL for Model 1, as presented in Table 4, reveals no diagnostic issues, as the null hypotheses are accepted across all tests (including serial correlation, normality, heteroskedasticity, CUSUM, and CUSUMQ). Thus, it can be concluded that the model is stable.

Table 4: Diagnostic Tests Results

	Model 1	Model 2
Symmetry	Coefficients/Probability	Coefficients/Probability
F_{pss}	6.96	27.07
W_{LR}	1.2331(0.329)	22.1941(0.018)**
W_{SR}	9.5042(0.023)**	6.3186(0.079)*
$\chi^2(\text{Serial correlation})$	(0.408)	(0.388)
$\chi^2(\text{normality})$	(0.689)	(0.904)
$\chi^2(\text{heteroscedasticity})$	(0.283)	(0.112)
<i>CUSUM</i>	Stable	Stable
<i>CUSUMQ</i>	Stable	Stable

Notes: F_{pss} : The F statistic results from the Wald test for long-term cointegration.

W_{LR} : Wald statistics of long-term symmetry.

W_{SR} : Wald statistics of short-term symmetry.

***, ** indicate the probability of rejecting the null hypothesis at 5% and 10%.

Source: Based on the authors' own calculations.

Examining the long-term estimations for Model 2, which considers the effects of export concentration on carbon emissions as shown in Table 3, a positive shock in export concentration is inversely related to carbon emissions in Türkiye during the relevant period, leading to a reduction in emissions of 0.31%. Conversely, a negative shock is positively associated with emissions, increasing by 0.36%. In these circumstances, positive and negative shocks in export concentration during the long term in Türkiye from 1995 to 2018 decreased and increased carbon emissions, respectively. Evaluating the impact of other variables on carbon emissions, it is evident that while the logarithmic GDP per capita level is positively correlated in the long run, the EKC or inverted U-shaped hypothesis holds true, as its square is negative. Additionally, a 1% rise in income inequality, government size, and renewable energy corresponds to increases in carbon emissions of 0.98%, 2.34%, and 0.26%, respectively. Reviewing the short-term nonlinear ARDL results for Model 2 reveals that, unlike the long-term findings, one and two lag negative shocks in export concentration raise carbon emissions by 0.46% and 0.34%, respectively. Furthermore, since the levels and squares of logarithmic GDP per capita are negative and positive in the short term, it is significant that the EKC relationship is U-shaped.

Once more, in contrast to the long term, increases in renewable energy in the short term lead to a 0.20% reduction in carbon emissions.

Examining the results of the diagnostic tests concerning the nonlinear ARDL for Model 2, as depicted in Table 4, which outlines the effects of export concentration on carbon emissions, there are no diagnostic issues, as all null hypotheses are accepted in the diagnostic tests (serial correlation, normality, heteroscedasticity, CUSUM, and CUSUMQ), indicating that the model is stable.

5. Discussion

The finding that export diversification or increased demand for new and existing products in the markets (Sharma et al. 2021, pp. 61739) reduces carbon emissions was compared with selected empirical literature. Accordingly, this finding contradicted studies by Gozgor & Can (2016), Adewuyi and Awodumi (2016), Adamu et al. (2019), Fang et al. (2019), Mania (2020), Wang et al. (2020), Can et al. (2020, 2021), Iqbal et al. (2021), and Khan et al. (2021) but was consistent with studies by Liu et al. (2019), Shahzad et al. (2020), and Zafar et al. (2021). Additionally, the finding that a positive shock to export concentration reduces carbon emissions while a negative shock increases carbon emissions was also compared with selected empirical literature and aligned with studies by Apergis et al. (2018) and Sharma et al. (2021).

Theoretically, a proportional relationship between export diversification and carbon emissions is anticipated. In cases of poor diversification, carbon emissions rise as export baskets typically consist of traditional products that are energy-inefficient and carbon-intensive (Sharma et al., 2021, pp. 61743-44). Conversely, it has been demonstrated that enhancing export diversification for Türkiye during the relevant period can improve environmental quality. Carbon emissions decline as the quality of export products increases in developing countries, as the export basket initially includes pollution-intensive goods. However, as the economy develops, these countries eliminate such products from their export basket, leading them to “import pollution-intensive goods from underdeveloped nations with less stringent environmental protection laws” (Gozgor and Can 2017, p. 2869).

Regarding export product concentration, a high concentration or the development of the product basket reduces carbon emissions because these products will increasingly be produced in developed countries. Ultimately, countries must specialize in export-grade products to lower carbon emissions. Export product concentration plays a crucial role in managing environmental pollution. Companies in developed countries should avoid producing goods

with high carbon emissions and instead aim to import them from developing nations. A carbon tax could be implemented to deter the production of carbon-intensive goods.

Additionally, exports can be expanded through learning by creating baskets. This necessitates the development of the financial sector and progress in foreign trade liberalization. Under these conditions, investors can enhance their export baskets through research, development, and effective regulations in the credit and labor markets (Apergis et al., 2018, pp. 14114–15). Another issue is that the negative impact of export diversification on carbon emissions lessens as the level of ecological innovation in the economy rises (Wang et al., 2020).

In the long term, the EKC hypothesis remains valid for Türkiye during the 1995-2018 period, showing that carbon emissions increase in parallel with income inequality, government size, and renewable energy growth. A comparison of results for the relevant control variables in both models studied against selected empirical literature indicates that the validity of the EKC hypothesis aligns with the findings of Ang (2007), Tamazian et al. (2009), Clement and Meunie (2010), Jalil and Feridun (2011), Pao and Tsai (2011), Iwata et al. (2012), Jayanthakumaran et al. (2012), Ozturk and Acaravcı (2013), Shahbaz et al. (2013b), Lau et al. (2014), Ling et al. (2015), Apergis (2016), Bilgili et al. (2016), Dogan and Seker (2016), Gozgor and Can (2016), Hübler (2017), Walde-Rufael and Idawu (2017), Zhan et al. (2017), Apergis et al. (2018), Katırcıoğlu and Katırcıoğlu (2018), Danish et al. (2019), Huang and Ucal (2019), Hailemariam et al. (2020), Can et al. (2020), Khan et al. (2021), and Sun et al. (2022). In other words, carbon emissions rise to a specific threshold as economic growth increases in Türkiye during the relevant period, but emissions decline as growth continues. Consequently, the inverted U-shaped hypothesis is valid for Türkiye. Thus, scale, technical, and composition effects are fully realized, indicating that the scale effect positively impacts carbon emissions beyond a threshold income level. In contrast, the technical effect has a negative influence. Therefore, increased economic activity enhances environmental quality and promotes cleaner production practices within the Turkish economy (Ling et al., 2015, p. 20137).

The findings of our study indicate that income inequality increases carbon emission-which contradicts the studies by Ravallion et al. (2000), Baek and Gweisah (2013), Hübler (2017), Safar (2022), Andersson (2023), Khan et al. (2023), and Wang, Yang, and Lie (2023). However, it aligns with research conducted by Clement and Meunie (2010), Zhang and Zhao (2014), Berthe and Elie (2015), Jorgenson (2015), Haoa et al. (2016), Jorgenson et al. (2017), Walde-Rufuel and Idawu (2017), Demir et al. (2019), Hailemariam et al. (2020), Yameogo and Dauda (2020), Ali (2021), Sharma et al. (2021), Ajide and Ibrahim (2022), Baloch and Danish (2022), Yang et al. (2022), and Wang et al. (2023). In other words, the notion that greater income inequality leads to increased carbon emissions and environmental harm can be explained

through economic and political channels. The economic channel relates to environmental degradation through the economic behavior of individuals, while the political channel stems from public policies aimed at environmental protection (Berthe and Elie, 2015). Therefore, reducing carbon emissions in Türkiye during the relevant period also depends on addressing income inequality.

A theoretical assessment of the finding that income inequality creates environmental pollution through increased carbon emissions in Türkiye during the specified period suggests that low-income levels hinder purchasing carbon-intensive goods. Meanwhile, high-income households exhibit a strong demand for such products. As a result, carbon emissions per capita rise as income shifts from the poor to the rich, driven by an increase in energy- or carbon-intensive goods and services. Consequently, the EKC shifts upward when income inequality is pronounced (Hübler, 2017, p. 176). This phenomenon is attributed to the Veblen effect and political economic arguments, where people with low incomes consume more energy and purchase carbon-intensive products as they progress toward the middle class (Jorgenson et al., 2017; Ali, 2021). Carbon emissions tend to decrease with economic growth as nations develop, and income equality improves since the affluent are more inclined to prioritize environmental protection and renewable energy usage. Another issue is that the influence of political elites diminishes over environmental policies in this pattern of income distribution. However, in nations with income inequality, carbon emissions may rise as societies opt for cheaper energy alternatives instead of adopting a greener lifestyle (Hailemariam et al., 2020; Sharma et al., 2021; Baloch and Danish, 2022).

The finding that government size increases carbon emissions contradicts studies by Halkos and Paizanos (2016), Zhan et al. (2017), Katircioglu and Katircioglu (2018), and Azam et al. (2023). However, it aligns with studies by Halkos and Paizanos (2013), Yameogo and Dauda (2020), Chen (2022), and Lan et al. (2022). From a theoretical perspective, there is a strong indirect effect on revenue as government size increases. Consequently, carbon emissions and environmental pollution rise through indirect effects as income levels increase (Halkos and Paizanos, 2013). Türkiye underwent fiscal transformation due to budget deficits throughout the 1990s and adopted a stringent fiscal policy during the 2000s. Tax rates are crucial for energy efficiency and environmental protection under such conditions (Katircioğlu and Katircioğlu, 2018, p. 5617). Regarding policy implications, increasing government size should promote green investment, clean energy, and environmental protection. In such scenarios, the government should allocate more resources for greener economic activities (Lan et al., 2022, p. 47231).

The finding that renewable energy consumption increases carbon emissions and deteriorates environmental quality contradicts studies by Baek (2016), Bilgili et al. (2016), Dogan and Seker (2016), Cheng et al. (2019), Mahmood et al. (2019), Leitao and Lorente (2020), Rahman and Vu (2020), Wang et al. (2020), Khan et al. (2021), Namahoro et al. (2021), Zafar et al. (2021), Wang et al. (2021), Adebayo et al. (2022), Akadiri and Adebayo (2022), Baloch and Danish (2022), Mushed and Dao (2022), Nkwatoh (2022), Raihan et al. (2022), Sun et al. (2022), Wang (2022), Adebayo et al. (2023), and Khan et al. (2023), but aligns with Demir et al. (2019). Türkiye should enhance resources for renewable energy, and decision-makers should implement energy-saving technologies. With these measures, carbon emissions will decrease, leading to environmental sustainability (Dogan and Seker, 2016; Akadiri and Adebayo, 2022).

6. Conclusion Remarks

This study considers the effects of various control variables (income inequality, government size, and renewable energy) in Türkiye from 1995 to 2018 to measure the asymmetric impacts of export diversification and concentration on carbon emissions. While the study's primary findings show that export diversification reduces carbon emissions and enhances environmental quality, this conclusion is corroborated by the result that a positive increase in export concentration also decreases carbon emissions. Additionally, both models found increased control variables to raise carbon emissions and environmental pollution.

Türkiye has maintained an outward-looking perspective since the 1980s, with export product diversity increasing alongside liberalization policies. However, incorporating new products into the export basket has resulted in transportation challenges, leading to higher energy consumption and carbon emissions (Can et al., 2021). This study has concluded that long-term export diversification reduces energy demand, as companies tend to avoid producing carbon-intensive goods, thus lowering carbon emissions. Türkiye needs to import carbon-intensive products while promoting renewable energy to mitigate emissions and environmental pollution (Gozgor and Can, 2016; Apergis, 2016; Can et al., 2020; Sharma et al., 2021; Akadiri and Adebayo, 2022). In other words, renewable energy should replace conventional energy through the composition effect, supported by the technical effect (Ling et al., 2015). Adopting environmentally friendly technologies through training is essential for reducing carbon emissions and enhancing awareness among decision-makers regarding energy security, efficiency, and environmental sustainability. Furthermore, the influence of export diversification on reducing carbon emissions is significant in encouraging green technologies and fostering innovation and technological development (Mahmood et al., 2019; Can et al., 2020; Shahzad et al., 2020; Sun et al., 2022).

On the other hand, the potential adverse effects of export diversification on carbon emissions can be further mitigated by increasing the presence of environmentally friendly products in the country's export basket (Khan et al., 2021). Moreover, lowering environmental pollution in an economy relies on government policies, the effective functioning of social institutions and markets, and improved governance (Kaika and Zervas, 2013a). In addition, Türkiye should aim to decrease its imports of pollution-intensive technologies and fossil fuels. Other financial instruments, such as green bonds, credit, and financing systems, can facilitate the transition to a low-carbon economy, along with subsidies for green investments (Khan et al., 2021; Wang, 2022). In conclusion, a structural transformation and updated manufacturing processes should be integrated into the design of foreign trade, environment, and energy policies, considering research and development investments that will enhance export quality (Wang et al., 2021; Murshed and Dao, 2022; Adebayo et al., 2023).

Our results are compatible with the UN's Sustainable Development Goals (SDGs). Therefore, according to SDG 7, Türkiye should increase its share of renewable energy in its portfolio by supporting renewable energy projects and related R&D activities. According to SDG 9, adopting clean, environmentally resilient technologies and industrial processes is essential, increasing the added value of industrial products, enhancing industrial diversity, boosting domestic R&D activities and funding, fostering innovation, and improving access to Information and Communication Technologies (ICTs). Additionally, support should be provided for technological development, resilient and sustainable infrastructure, and financial access for companies and small and medium enterprises (SMEs) while ensuring their integration into market and value chains. Lastly, as decreasing carbon emissions is one of the pillars of achieving climate change goals based on SDG 13, our findings emphasize the crucial role of emission reduction efforts in climate change mitigation and adaptation. Türkiye should focus on capacity building for climate change adaptation, integrating climate change-related strategies into policies, plans, and other strategies, raising awareness related to climate change through educational activities, and enhancing institutional and individual capacity building. Furthermore, mechanisms should be established to improve management and effective planning capacities related to climate change (United Nations 2020, 2023).

References

- Adewuyi, O. A., Awodumi, B. O. (2016). Analysis of the environmental pollution effect of Nigeria's export diversification drive. *Annual Conference of the Nigerian Economic Society*, Abuja, Nigeria, from August 15–18, 1–26.
- Adamu, T. M., Haq, I., Shafiq, M. (2019). Analyzing the impact of energy, export variety, and FDI on environmental degradation in the context of Environmental Kuznets Curve hypothesis: A case study of India. *Energies*, 12(6), 1076, 1-18. <https://doi.org/10.3390/en12061076>
- Adebayo, T. S., Awosusi, A. A., Rjoub, H., Agyekum, E. B., Kırıkkaleli, D. (2022). The influence of renewable energy usage on consumption-based carbon emissions in MINT countries. *Heliyon*, 8(e08941), 1-11. <https://doi.org/10.1016/j.heliyon.2022.e08941>
- Adebayo, T. S., Rjoub, H., Akinsola, G. D., Oladipupo, S. D. (2022). The asymmetric effects of renewable energy consumption and trade openness on carbon emissions in Sweden: New evidence from quantile-on-quantile regression approach. *Environmental Science and Pollution Research*, 29, 1875-1886. <https://doi.org/10.1007/s11356-021-15706-4>
- Adebayo, T. S., Oladipupo, S. D., Rjoub, H., Kırıkkaleli, D., Adeshola, I. (2023). Asymmetric effect of structural change and renewable energy consumption on carbon emissions: Designing an SDG framework for Türkiye. *Environment, Development and Sustainability*, 25, 528-556. <https://doi.org/10.1007/s10668-021-02065-w>
- Ajide, B., Ibrahim, L. (2022). Environmental impacts of income inequality: Evidence from G7 economies. *Environmental Science and Pollution Research*, 29, 1887-1908. <https://doi.org/10.1007/s11356-021-15720-6>
- Akadiri, S. S., Adebayo, T. S. (2022). Asymmetric nexus among financial globalization, non-renewable energy, renewable energy use, economic growth, and carbon emissions impact on environmental sustainability targets in India. *Environmental Science and Pollution Research*, 29, 16311-16323. <https://doi.org/10.1007/s11356-021-16849-0>
- Ali, I. M. A. (2022). Income inequality and environmental degradation in Egypt: Evidence from dynamic ARDL approach. *Environmental Science and Pollution Research*, 29, 8408-8422. <https://doi.org/10.1007/s11356-021-16275-2>
- Andersson, F. N. G. (2023). Income inequality and carbon emissions in the United States 1929–2019. *Ecological Economics*, 204, 107633, 1-14 . <https://doi.org/10.1016/j.ecolecon.2022.107633>
- Apergis, N. (2016). Environmental Kuznets curves: New evidence on both panel and country-level CO2 emissions. *Energy Economics*, 54, 263-271. <https://doi.org/10.1016/j.eneco.2015.12.007>
- Apergis, N., Can, M., Gozgor, G., Lau, C. M. (2018). Effects of export concentration on CO2 emissions in developed countries: An empirical analysis. *Environmental Science and Pollution Research*, 25, 14106-14116. <https://doi.org/10.1007/s11356-018-1634-x>
- Ang, J. B. (2007). CO2 emissions, energy consumption, and output in France. *Energy Policy*, 35, 4772-4778. <https://doi.org/10.1016/j.enpol.2007.03.032>

- Ali, R., Alwang, J., Siegel, B. P. (1991). Is export diversification the best way to achieve export growth and stability: A look at three African countries. *Policy, Research, and External Affairs Working Papers No. WPS 729*, Agriculture Operations, Washington, D.C.: World Bank Group. [Retrieved 2023-08-01] Available at: <http://documents.worldbank.org/curated/en/635281468774612514/Is-export-diversification-the-best-way-to-achieve-export-growth-and-stability-A-look-at-three-African-countries>
- Azam, W., Khan, I., Ali, S. A. (2023). Alternative energy and natural resources in determining environmental sustainability: A look at the role of government final consumption expenditures in France. *Environmental Science and Pollution Research*, 30, 1949-1965. <https://doi.org/10.1007/s11356-022-22334-z>
- Baek, J. (2016). Do nuclear and renewable energy improve the environment: Evidence from the United States. *Ecological Indicators*, 66, 352-356. <https://doi.org/10.1016/j.ecolind.2016.01.059>
- Baek, J., Gweisah, G. (2013). Does income inequality harm the environment?: Empirical evidence from the United States. *Energy Policy*, 62, 1434-1437. <https://doi.org/10.1016/j.enpol.2013.07.097>
- Baloch, M. A., Danish. (2022). The nexus between renewable energy, income inequality, and consumption-based CO2 emissions: An empirical investigation. *Sustainable Development*, 30(5), 1268-1277. <https://doi.org/10.1002/sd.2315>
- Bekhet, H. A., Matar, A., Yasmin, T. (2017). CO2 emissions, energy consumption, economic growth, and financial development in GCC countries: Dynamic simultaneous equation models. *Renewable and Sustainable Energy Reviews*, 70, 117-132. <https://doi.org/10.1016/j.rser.2016.11.089>
- Berthe, A., Elie, L. (2015). Mechanisms explaining the impact of economic inequality on environmental deterioration. *Ecological Economics*, 116, 191-200. <https://doi.org/10.1016/j.ecolecon.2015.04.026>
- Berthelemy, J. C., Chauvin, S. (2000). *Structural changes in Asia and growth prospects after the crisis. CEPII Working Papers*.
- Bilgili, F., Koçak, E., Bulut, Ü. (2016). The dynamic impact of renewable energy consumption on CO2 emissions: A revisited environmental Kuznets curve approach. *Renewable and Sustainable Energy Reviews*, 54, 838-845. <https://doi.org/10.1016/j.rser.2015.10.080>
- Cadot, O., Carrere, C., Strauss-Kahn, V. (2011). Export diversification: What's behind the hump? *Review of Economics and Statistics*, 93(2), 590-605. https://doi.org/10.1162/REST_a_00078
- Can, M., Gözgör, G. (2017). The impact of economic complexity on carbon emissions: Evidence from France. *Environmental Science and Pollution Research*, 24, 16364-16370. <https://doi.org/10.1007/s11356-017-9219-7>
- Can, M., Ahmad, M., Khan, Z. (2021). The impact of export composition on environment and energy demand: Evidence from newly industrialized countries. *Environmental Science and Pollution Research*, 28, 33599-33612. <https://doi.org/10.1007/s11356-021-13084-5>

- Can, M., Dogan, B., Saboori, B. (2020). Does trade matter for environmental degradation in developing countries? New evidence in the context of export product diversification. *Environmental Science and Pollution Research*, 27, 14702-14710. <https://doi.org/10.1007/s11356-020-08000-2>
- Cao, G., Xie, F., Ling, M. (2022). Spillover effects in Chinese carbon, energy and financial markets. *International Finance*, 25(3), 416-434. <https://doi.org/10.1111/infi.12417>
- Chen, L. (2022). How CO2 emissions respond to changes in government size and level of digitalization? Evidence from the BRICS countries. *Environmental Science and Pollution Research*, 29, 457-467. <https://doi.org/10.1007/s11356-021-15693-6>
- Cheng, C., Ren, X., Wang, Z., Yan, C. (2019). Heterogeneous impacts of renewable energy and environmental patents on CO2 emission - Evidence from the BRIICS. *Science of the Total Environment*, 668, 1328-1338. <https://doi.org/10.1016/j.scitotenv.2019.02.063>
- Clement, M., Meunie, A. (2010). Is inequality harmful for the environment? An empirical analysis applied to developing and transition countries. *Review of Social Economy*, 68(4), 413-444. <https://doi.org/10.1080/00346760903480590>
- Danisha, Baloch, M. A., Wang, B. (2019). Analyzing the role of governance in CO2 emissions mitigation: The BRICS experience. *Structural Change and Economic Dynamics*, 51, 119-125. <https://doi.org/10.1016/j.strueco.2019.08.007>
- Demir, C., Cergibozan, R., Gök, A. (2019). Income inequality and CO2 emissions: Empirical evidence from Türkiye. *Energy and Environment*, 30(3), 444-461. <https://doi.org/10.1177/0958305X18793109>
- Doğan, E., Şeker, F. (2016). Determinants of CO2 emissions in the European Union: The role of renewable and non-renewable energy. *Renewable Energy*, 94, 429-439. <https://doi.org/10.1016/j.renene.2016.03.078>
- Doğan, B., Mara, M., Kumar, A. T., Shawkat, H. (2020). Impacts of export quality on environmental degradation: Does income matter? *Environmental Science and Pollution Research*, 27, 13735-13772. <https://doi.org/10.1007/s11356-019-07371-5>
- Erdogan, E. Serin Oktay, D., Manga, M., Bal, H., Algan, N. (2024) "Examining the effects of renewable energy and economic growth on carbon emission in Canada: evidence from the nonlinear ARDL approaches. *Evaluation Review*, 48(1), 63-89. <https://doi.org/10.1177/0193841X231166973>
- Fang, J., Gözgör, G., Lu, Z., Wu, W. (2019). Effects of the export product quality on carbon dioxide emissions: Evidence from developing economies. *Environmental Science and Pollution Research*, 26, 12181-12193. <https://doi.org/10.1007/s11356-019-04513-7>
- Gözgör, G., Can, M. (2016). Export product diversification and the Environmental Kuznets Curve: Evidence from Türkiye. *Environmental Science and Pollution Research*, 23, 21594-21603. <https://doi.org/10.1007/s11356-016-7403-9>

- Gözcör, G., Can, M. (2017). Does export product quality matter for CO₂ emissions? Evidence from China. *Environmental Science and Pollution Research*, 24, 2866-2875.
<https://doi.org/10.1007/s11356-016-8070-6>
- Grossman, G. M., Krueger, A. B. (1995). Economic growth and the environment. *Quarterly Journal of Economics*, 110, 353-377. <https://doi.org/10.2307/2118443>
- Hailemariam, A., Dzhumashev, R., Shahbaz, M. (2020). Carbon emissions, income inequality, and economic development. *Empirical Economics*, 59, 1139-1159.
<https://doi.org/10.1007/s00181-019-01664-x>
- Halkos, G. E., Paizanos, E. A. (2013). The effect of government expenditure on the environment: An empirical investigation. *Ecological Economics*, 91, 48-56.
<https://doi.org/10.1016/j.ecolecon.2013.04.002>
- Halkos, G. E., Paizanos, E. A. (2016). The effects of fiscal policy on CO₂ emissions: Evidence from the U.S.A. *Energy Policy*, 88, 317-328. <https://doi.org/10.1016/j.enpol.2015.10.035>
- Hao, Y., Chen, H., Zhang, Q. (2016). Will income inequality affect environmental quality? Analysis based on China's provincial panel data. *Ecological Indicators*, 67, 533-542.
<http://dx.doi.org/10.1016/j.ecolind.2016.03.025>
- Haug, A.A., & Ucal, M. (2019). The role of trade and FDI for CO₂ emissions in Türkiye: Nonlinear relationships. *Energy Economics*, 81, 297-307. <https://doi.org/10.1016/j.eneco.2019.04.006>
- Hübler, M. (2017). The inequality-emissions nexus in the context of trade and development: A quantile regression approach. *Ecological Economics*, 134, 174-185.
<https://doi.org/10.1016/j.ecolecon.2016.12.015>
- Iqbal, N., Abbasi, K. R., Shinwari, R., Guangcai, W., Ahmad, M., & Tang, K. (2021). Does exports diversification and environmental innovation achieve carbon neutrality target of OECD economies? *Journal of Environmental Management*, 291, 112648, 1-10.
<https://doi.org/10.1016/j.jenvman.2021.112648>
- Jalil, A., & Feridun, M. (2011). The impact of growth, energy and financial development on the environment in China: a cointegration analysis. *Energy Economics*, 33, 284-291.
<https://doi.org/10.1016/j.eneco.2010.10.003>
- Jayanthakumaran, K., Verma, R., & Liu, Y. (2012). CO₂ emissions, energy consumption, trade and income: a comparative analysis of China and India. *Energy Policy*, 42, 450-460.
<https://doi.org/10.1016/j.enpol.2011.12.010>
- Jorgenson, A.K. (2015). Inequality and the carbon intensity of human well-being. *Journal of Environmental Studies and Sciences*, 5, 277-282. <https://doi.org/10.1007/s13412-015-0234-z>
- Jorgenson, A., Schor, J., Huang, X. (2017). Income inequality and carbon emissions in the United States: A state-level analysis, 1997-2012. *Ecological Economics*, 134, 40-48.
<https://doi.org/10.1016/j.ecolecon.2016.12.016>
- Kaika, D., Zervas, E. (2013a). The environmental Kuznets curve (EKC) theory - Part A: Concept, causes, and the CO₂ emissions case. *Energy Policy*, 62, 1392-1402.
<https://doi.org/10.1016/j.enpol.2013.07.131>

- Kaika, D., Zervas, E. (2013b). The environmental Kuznets curve (EKC) theory - Part B: Critical issues. *Energy Policy*, 62, 1403-1411. <https://doi.org/10.1016/j.enpol.2013.07.130>
- Katircioğlu, S., Katircioğlu, S. (2018). Testing the role of fiscal policy in environmental degradation: The case of Türkiye. *Environmental Science and Pollution Research*, 25, 5616-5630. <https://doi.org/10.1007/s11356-017-0906-1>
- Khan, I., Han, L., Zhong, R., Bibi, R., Khan, H. (2023). Income inequality, economic growth, renewable energy usage, and environmental degradation in the Belt and Road initiative countries: Dynamic panel estimation. *Environmental Science and Pollution Research*, 30, 57142-57154. <https://link.springer.com/article/10.1007/s11356-023-26273-1>
- Khan, Z., Murshed, M., Dong, K., Yang, S. (2021). The roles of export diversification and composite country risks in carbon emissions abatement: Evidence from the signatories of the Regional Comprehensive Economic Partnership agreement. *Applied Economics*, 53 (41), 4769-4787. <https://doi.org/10.1080/00036846.2021.1907289>
- Kwiatkowski, D., Phillips, P., Schmidt, P., Shin, Y. (1992). Testing the null hypothesis stationary against the alternative of a unit root: How sure are we that economic time series have a unit root? *Journal of Econometrics*, 54, 159-187.
- Kuznets, S. (1955). Economic growth and income inequality. *American Economic Review*, 45, 1-28.
- Iwata, H., Okada, K., Samwoth, S. (2012). Empirical study on the determinants of CO2 emissions: Evidence from OECD countries. *Applied Economics*, 44(27), 3513-3519. <https://doi.org/10.1080/00036846.2011.577023>
- Lan, H., Cheng, C., Sohail, M. T. (2022). Asymmetric determinants of CO2 emissions in China: Do government size and economic size matter?. *Environmental Science and Pollution Research*, 29, 47225-47232. <https://doi.org/10.1007/s11356-022-19096-z>
- Lau, L. S., Choong, C. K., Eng, Y. K. (2014). Investigation of the environmental Kuznets curve for carbon emissions in Malaysia: Do foreign direct investment and trade matter?. *Energy Policy*, 68, 490-497. <https://doi.org/10.1016/j.enpol.2014.01.002>
- Leitão, N. C., Lorente, D. B. (2020). The linkage between economic growth, renewable energy, tourism, CO2 emissions, and international trade: The evidence for the European Union. *Energies*, 13(2020), 4838, 1-16. <https://doi.org/10.3390/en13184838>
- Lind, J. T., Mehlum, H. (2010). With or without U? The appropriate test for a U-shaped relationship. *Oxford Bulletin of Economics and Statistics*, 72(1), 109-118. <https://doi.org/10.1111/j.1468-0084.2009.00569.x>
- Liu, H., Kim, H., Choe, J. (2019). Export diversification, CO2 emissions, and EKC: Panel data analysis of 125 countries. *Asia-Pacific Journal of Regional Science*, 3, 361-393. <https://doi.org/10.1007/s41685-018-0099-8>
- Ling, C. H., Ahmed, K., Binti, M. R., Shahbaz, M. (2015). Decomposing the trade-environment nexus for Malaysia: What do the technique, scale, composition, and comparative advantage effects indicate? *Environmental Science and Pollution Research*, 22, 20131-20142. <https://doi.org/10.1007/s11356-015-5217-9>

- Mahmood, N., Wang, Z., Hassan, S. T. (2019). Renewable energy, economic growth, human capital, and CO₂ emission: An empirical analysis. *Environmental Science and Pollution Research*, 26, 20619-20630. <https://doi.org/10.1007/s11356-019-05387-5>
- Mania, E. (2019). Export diversification and CO₂ emissions: An augmented environmental Kuznets curve. *Journal of International Development*, 3, 168-185. <https://doi.org/10.1002/jid.3441>
- Muhammad, I., Ozcan, R., Jain, V., Sharma, P., Shabbir, M. S. (2022). Does environmental sustainability affect the renewable energy consumption? Nexus among trade openness, CO₂ emissions, income inequality, renewable energy, and economic growth in OECD countries. *Environmental Science and Pollution Research*, 29, 90147-90157. <https://doi.org/10.1007/s11356-022-22011-1>
- Murshed, M., Dao, N. T. T. (2022). Revisiting the CO₂ emission-induced EKC hypothesis in South Asia: The role of export quality improvement. *GeoJournal*, 87, 535-563. <https://doi.org/10.1007/s10708-020-10270-9>
- Namahoro, J. P., Wu, Q., Zhou, N., Xue, S. (2021). Impact of energy intensity, renewable energy, and economic growth on CO₂ emissions: Evidence from Africa across regions and income levels. *Renewable and Sustainable Energy Reviews*, 147(111233), 1-13. <https://doi.org/10.1016/j.rser.2021.111233>
- Nasir, M., Rehman, F. U. (2011). Environmental Kuznets curve for carbon emissions in Pakistan: An empirical investigation. *Energy Policy*, 39, 1857-1864. <https://doi.org/10.1016/j.enpol.2011.01.025>
- Narayan, P. K. (2005). The saving and investment nexus for China: Evidence from cointegration tests. *Applied Economics*, 37(17), 1979-1990. <https://doi.org/10.1080/00036840500278103>
- Nkwatoh, L. S. (2022). Zero-pollution effect and economic development: Standard and nested environmental Kuznets curve analyses for West Africa. *Environmental Development and Sustainability*, 24, 11895-11910. <https://doi.org/10.1007/s10668-021-01921-z>
- Oster, E. (2019). Unobservable Selection and Coefficient Stability: Theory and Evidence. *Journal of Business & Economic Statistics*, 37(2), 187-204. <https://doi.org/10.1080/07350015.2016.1227711>
- Öztürk, İ., Acaravcı, A. (2013). The long run and causal analysis of energy, growth, openness and financial development on carbon emissions in Türkiye. *Energy Economics*, 36, 262-267. <https://doi.org/10.1016/j.eneco.2012.08.025>
- Papageorgiou, C., Spatafora, N. (2012). Economic diversification in LICs: Stylized facts and macroeconomic implications. *IMF Staff Discussion Note*, 14, 1-19.
- Pao, H. J., Tsai, C. M. (2011). Multivariate Granger causality between CO₂ emissions, energy consumption, FDI, and GDP: Evidence from a panel of BRIC countries. *Energy*, 36, 685-693. <https://doi.org/10.1016/j.energy.2010.09.041>
- Pesaran, M. H., Shin, Y., Smith, R. J. (2001). Bounds testing approaches to the analysis of level relationships. *Journal of Applied Econometrics*, 16, 289-326. <https://doi.org/10.1002/jae.616>
- Phillips, P. C. B., Perron, P. (1988). Testing for a unit root in time series regression. *Biometrika*, 75, 335-346. <https://doi.org/10.2307/2336182>

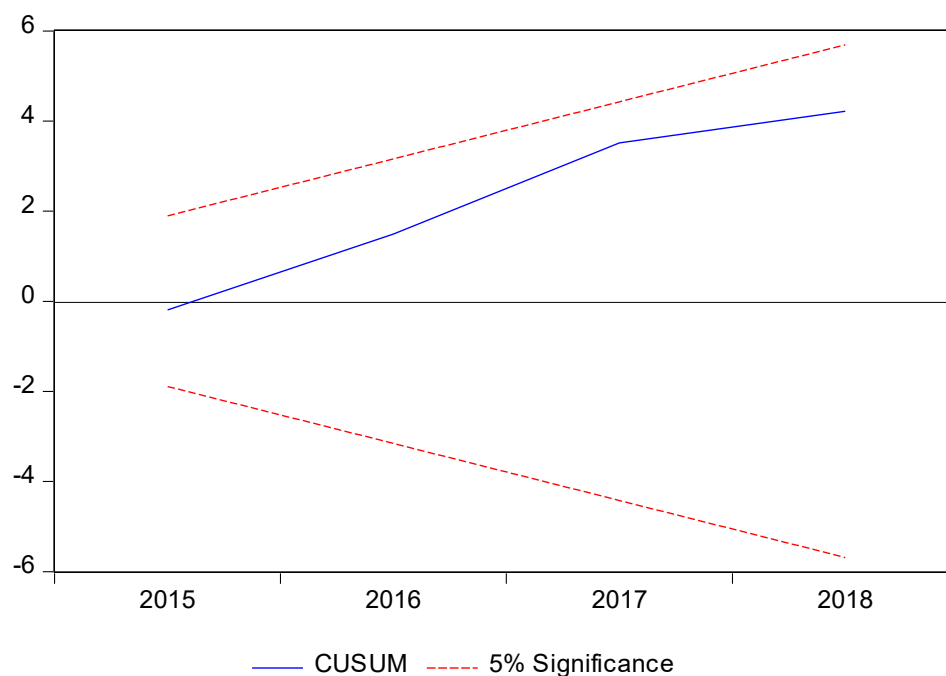
- Rahman, M. M., Vu, X.-B. (2020). The nexus between renewable energy, economic growth, trade, urbanisation and environmental quality: A comparative study for Australia and Canada. *Renewable Energy*, 155, 617-627. <https://doi.org/10.1016/j.renene.2020.03.135>
- Raihan, A., Begum, R. A., Said, M. N. M., Pereira, J. J. (2022). Relationship between economic growth, renewable energy use, technological innovation, and carbon emission toward achieving Malaysia's Paris agreement. *Environmental Systems Decisions*, 42, 586-607. <https://doi.org/10.1007/s10669-022-09848-0>
- Ravallion, M., Heil, M., Jalan, J. (2000). Carbon emissions and income inequality. *Oxford Economic Papers*, 52, 651-669. <https://doi.org/10.1093/oep/52.4.651>
- Safar, W. (2022). Income inequality and CO2 emissions in France: Does income inequality indicator matter?. *Journal of Cleaner Production*, 370, 1-27. 133457. <https://doi.org/10.1016/j.jclepro.2022.133457>
- Sharma, R., Sinha, A., Kautish, P. (2021). Examining the nexus between export diversification and environmental pollution: Evidence from BRICS nations. *Environmental Science and Pollution Research*, 28, 61732-61747. <https://doi.org/10.1007/s11356-021-14889-0>
- Shahbaz, M., Hye, Q. M. A., Tiwari, A. K., Leitao, N. C. (2013a). Economic growth, energy consumption, financial development, international trade, and CO2 emissions in Indonesia. *Renewable and Sustainable Energy Reviews*, 25, 109-121. <https://doi.org/10.1016/j.rser.2013.04.009>
- Shahbaz, M., Tiwari, A. K., Nasir, M. (2013b). The effects of financial development, economic growth, coal consumption, and trade openness on CO2 emissions in South Africa. *Energy Policy*, 61, 1452-1459. <https://doi.org/10.1016/j.enpol.2013.07.006>
- Shahzad, U., Ferraz, D., Dogan, B., Rebelatto, D. A. D. N. (2020). Export product diversification and CO2 emissions: Contextual evidences from developing and developed economies. *Journal of Cleaner Production*, 276, 1-19. 124146. <https://doi.org/10.1016/j.jclepro.2020.124146>
- Shin, Y., Yu, B., Greenwood-Nimmo, M. (2014). Modelling asymmetric cointegration and dynamic multipliers in a nonlinear ARDL framework. In R. C. Sickles & W. C. Horrace (Eds.), *Festschrift in honor of Peter Schmidt: Econometric methods and applications* (pp. 281-314). Springer. https://doi.org/10.1007/978-1-4899-8008-3_9
- Subramaniam, Y., Loganathan, N. (2024). Does green finance affect renewable energy development in Singapore?. *Journal of Asian Business and Economic Studies*, 31(3), 162-174. <https://doi.org/10.1108/JABES-02-2023-0052>
- Sun, Y., Li, H., Andlib, Z., Genie, M. G. (2022). How do renewable energy and urbanization cause carbon emissions? Evidence from advanced panel estimation techniques. *Renewable Energy*, 185, 996-1005. <https://doi.org/10.1016/j.renene.2021.12.112>
- Tamazian, A., Chousa, J. P., Vadlamanati, K. C. (2009). Does higher economic and financial development lead to environmental degradation: Evidence from BRIC countries. *Energy Policy*, 37, 246-253. <https://doi.org/10.1016/j.enpol.2008.08.025>

- Uddin, M. M., Mishra, V., Smyth, R. (2020). Income inequality and CO2 emissions in the G7, 1870–2014: Evidence from non-parametric modelling. *Energy Economics*, 88, 104780, 1-14. <https://doi.org/10.1016/j.eneco.2020.104780>
- United Nations (2020). *The sustainable development goals report*. [Retrieved 2022-02-18] Available at: <https://unstats.un.org/sdgs/report/2020>.
- United Nations (2023) The 17 Goals - Sustainable development goals-the United Nations. [Retrieved 2023-08-01] Available at: <https://sdgs.un.org/goals>.
- Walde-Rufael, Y., Idowu, S. (2017). Income distribution and CO2 emission: A comparative analysis for China and India. *Renewable and Sustainable Energy Reviews*, 74, 1336-1345. <https://doi.org/10.1016/j.rser.2016.11.149>.
- Wang, Q., Li, L., Li, R. (2023). Uncovering the impact of income inequality and population aging on carbon emission efficiency: An empirical analysis of 139 countries. *Science of The Total Environment*, 857, 159508, 1-11. <https://doi.org/10.1016/j.scitotenv.2022.159508>.
- Wang, Q., Yang, T., Li, R. (2023). Does income inequality reshape the environmental Kuznets curve (EKC) hypothesis? A nonlinear panel data analysis. *Environmental Research*, 216, 114575, 1-15. <https://doi.org/10.1016/j.envres.2022.114575>.
- Wang, L., Chang, H-L., Rizvi, S.K.A., Sari, A. (2020). Are eco-innovation and export diversification mutually exclusive to control carbon emissions in G-7 countries? *Journal of Environmental Management*, 270, 110829, 1-8. <https://doi.org/10.1016/j.jenvman.2020.110829>.
- Wang, Z., Jebli, M.B., Madaleno, M., Dogan, B., Shahzad, U. (2021). Does export product quality and renewable energy induce carbon dioxide emissions: Evidence from leading complex and renewable energy economies. *Renewable Energy*, 171, 360-370. <https://doi.org/10.1016/j.renene.2021.02.066>.
- Wang, L. (2022). Research on the dynamic relationship between China's renewable energy consumption and carbon emissions based on ARDL model. *Resources Policy*, 77, 102764, 1-9. <https://doi.org/10.1016/j.resourpol.2022.102764>.
- World Bank Group (2022). Türkiye CCDR Background Note 7 The decarbonizing world: Competitive risks, strengths, and opportunities for Türkiye, Retrieved from <https://documents1.worldbank.org/curated/en/099005106072227043/pdf/P1774790c1683a00808054028e978b5878a.pdf>, Access Date: 31.03.2025
- Yameogo, C., Dauda, E. W., Risikat, O. S. (2020). The effect of income inequality and economic growth on environmental quality: A comparative analysis between Burkina Faso and Nigeria. *Journal of Public Affairs*, 2020: e2566, 1-15. <https://doi.org/10.1002/pa.2566>.
- Yang, G., Sun, T., Wang, J., Li, X. (2015). Modeling the nexus between carbon dioxide emissions and economic growth. *Energy Policy*, 86, 104-117. <https://doi.org/10.1016/j.enpol.2015.06.031>.
- Yang, B., Ali, M., Hashmi, S. H., Jahanger, A. (2022). Do income inequality and institutional quality affect CO2 emissions in developing economies? *Environmental Science and Pollution Research*, 29, 42720-42734. <https://doi.org/10.1007/s11356-021-18278-5>.

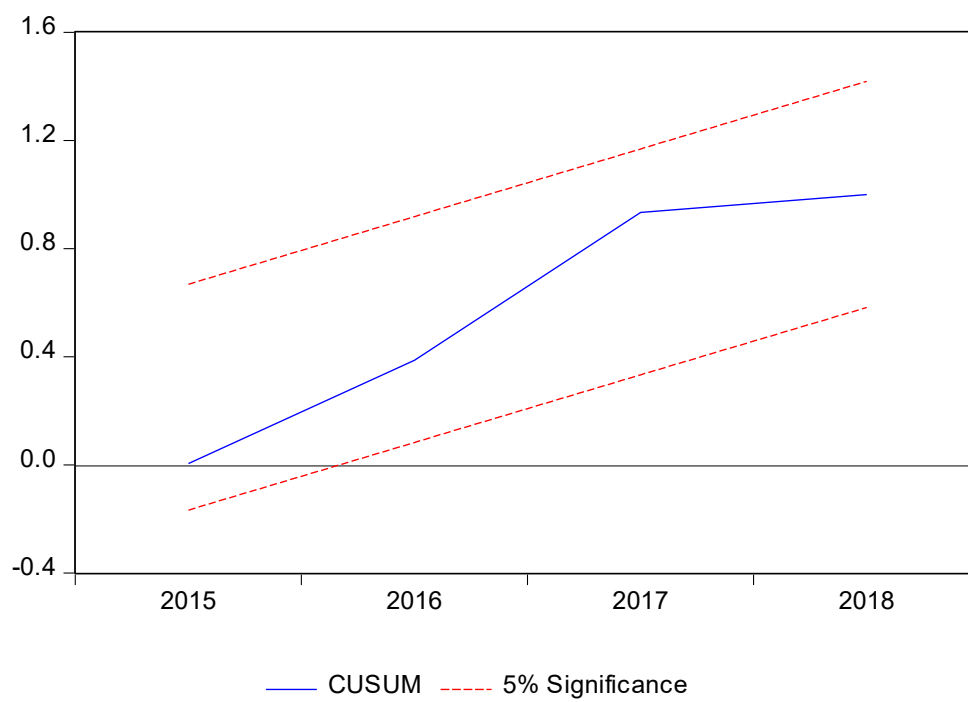
- Zafar, M. W., Saleem, M. M., Destek, M. A., Çağlar, A. E. (2021). The dynamic linkage between remittances, export diversification, education, renewable energy consumption, economic growth and CO2 emissions in top remittance-receiving countries. *Sustainable Development*, 30(1), 165-175. <https://doi.org/10.1002/sd.2236>.
- Zhang, Q., Zhang, S., Ding, Z., Hao, Y. (2017). Does government expenditure affect environmental quality? Empirical evidence using Chinese city-level data. *Journal of Cleaner Production*, 161, 143-152. <https://doi.org/10.1016/j.jclepro.2017.05.096>.
- Zhang, Y. J. (2011). The impact of financial development on carbon emissions: An empirical analysis in China. *Energy Policy*, 39, 2197-2203. <https://doi.org/10.1016/j.enpol.2011.02.026>.
- Zhang, C., & Zhao, W. (2014). Panel estimation for income inequality and CO2 emissions: A regional analysis in China. *Applied Energy*, 136, 382-392. <http://dx.doi.org/10.1016/j.apenergy.2014.09.048>

Appendix 1:

CUSUM and CUSUMQ for Model 1

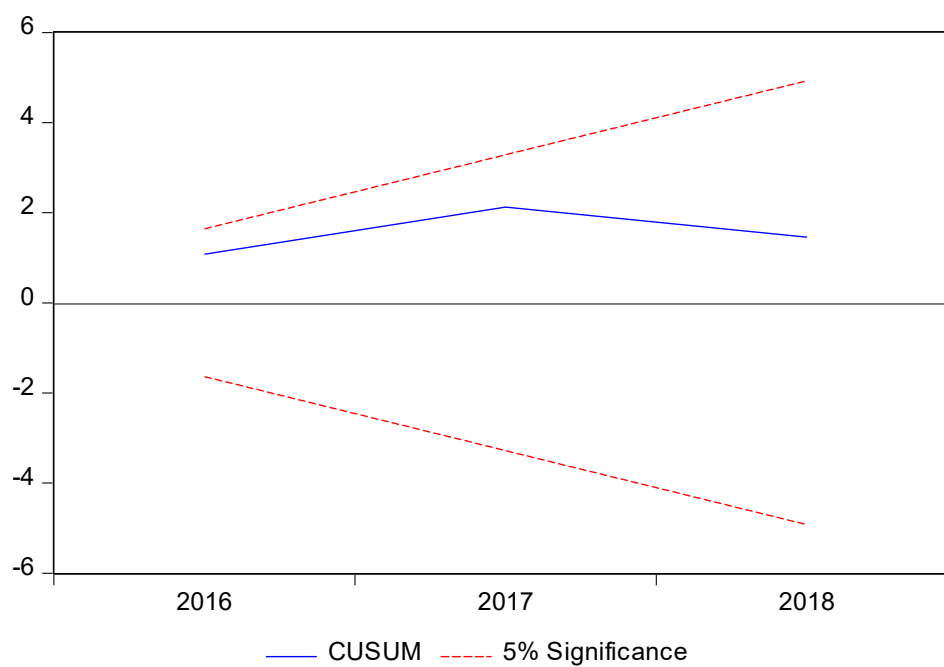


Source: Based on the authors' own calculations.

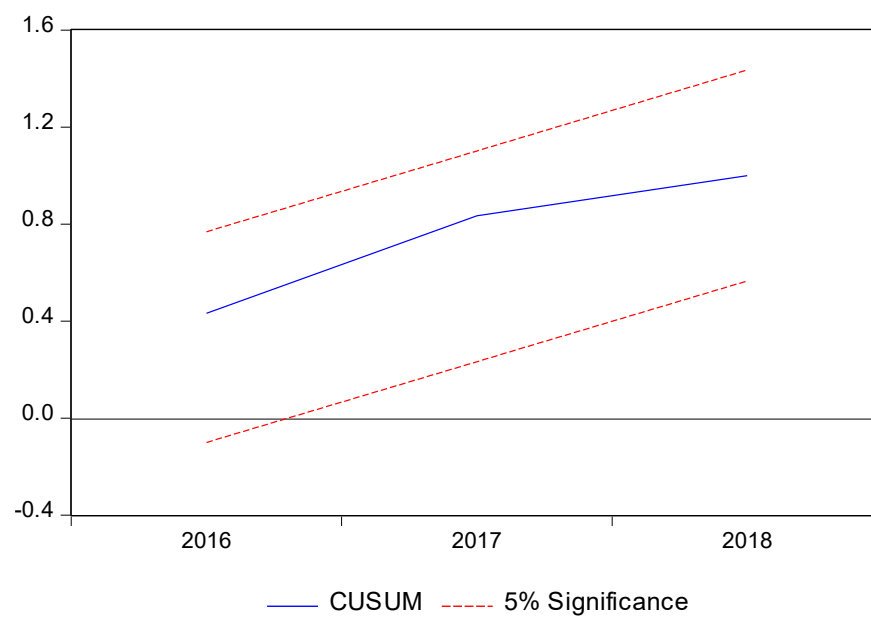


Source: Based on the authors' own calculations.

Appendix 2: CUSUM and CUSUMQ for Model 2



Source: Based on the authors' own calculations.



Source: Based on the authors' own calculations.